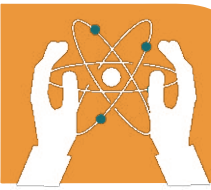


UNIT 1

LIVING THINGS

Learning objectives:

- Identify characteristics that distinguish living things from non-living things.
- Learn the classification of living organisms.
- Able to understand the adaptations in living organisms.
- Discuss the role of living organisms in environment.



Think Like a Scientist

If scientists discovered an unknown object in space, what characteristics would they need to observe to determine if it is a living organism?

Living things are distinct from non-living things because they exhibit certain characteristics, such as movement, growth, respiration, reproduction, response to stimuli, excretion, and adaptation. These characteristics ensure their survival and reproduction.

1.1 Characteristics of Living Things

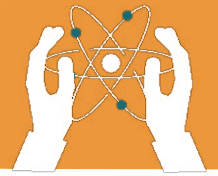
1. Growth: All living things grow over a time.

Example: A seed grows into a tree.

2. Reproduction: Living things produce offspring to ensure the continuation of their species.

Example: Offspring of humans, animal babies, birds laying eggs.

3. Movement: Animals move to find food and escape predators, while plants move their parts, such as



leaves turning towards light.

4. Respiration: Energy is obtained by breaking down food.

5. Response to Stimuli: Living things react to changes in their environment, such as light, heat, or danger.

6. Excretion: Waste materials like carbon dioxide, sweat, and urine are removed from the body.

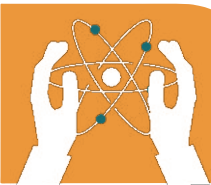
7. Adaptation: Organisms change over time to better survive in their environment.

1.2 Classification of Living Organisms

Living organisms are diverse and exist in various forms, from microscopic bacteria to giant trees and animals. To study them systematically, scientists classify organisms based on their similarities and differences. Classification helps in identifying, naming, and understanding the relationships between different organisms.

CORE INSIGHTS

- 1. Unicellular Organisms:** Single-celled organisms like bacteria are unicellular.
- 2. Multicellular Organisms:** Organisms with multiple cells, e.g., humans, animals, and plants are multicellular.
- 3. Cell Structure:** Includes a membrane, nucleus, and cytoplasm for essential functions.
- 4. Cell Functions:** Specialized roles, e.g., muscle cells for movement and nerve cells for signaling.



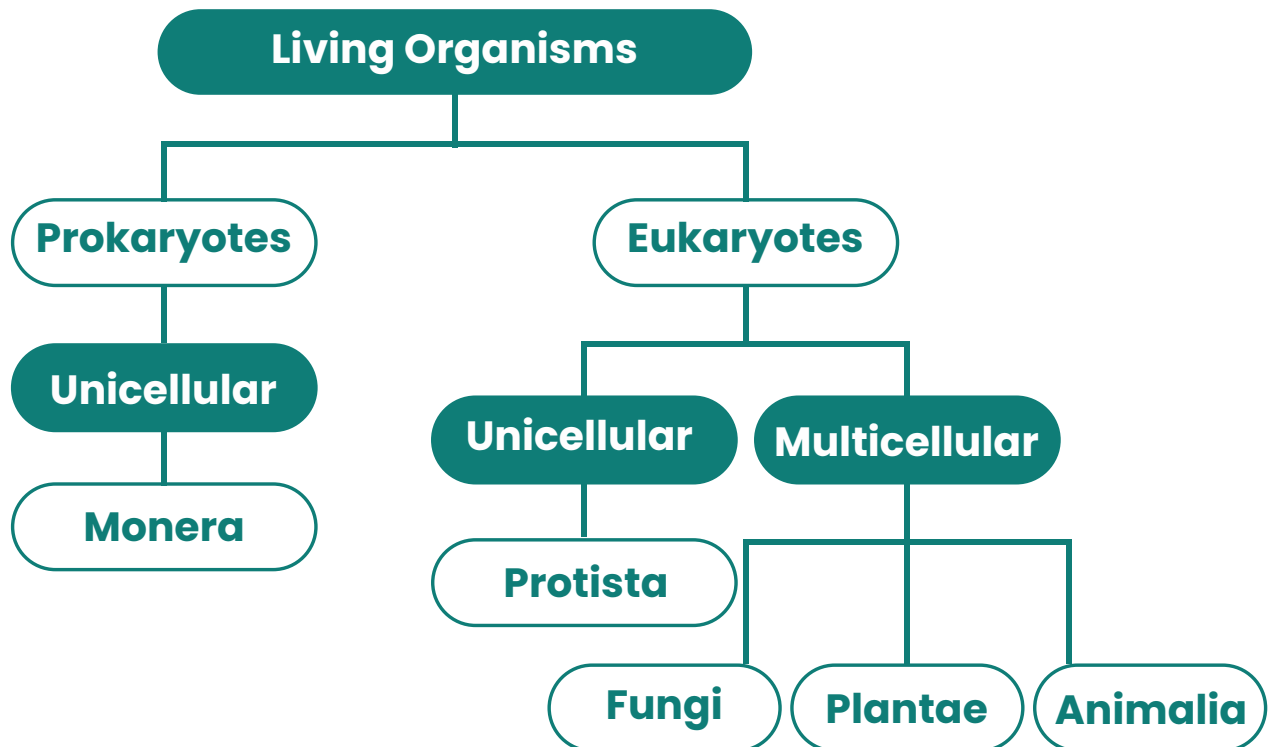
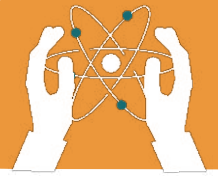
1.3 Importance of Classification

- It organizes living organisms into groups for easier study.
- It helps in understanding evolutionary relationships.
- It avoids confusion by providing a universal naming system (scientific names).

1.4 Basis of Classification

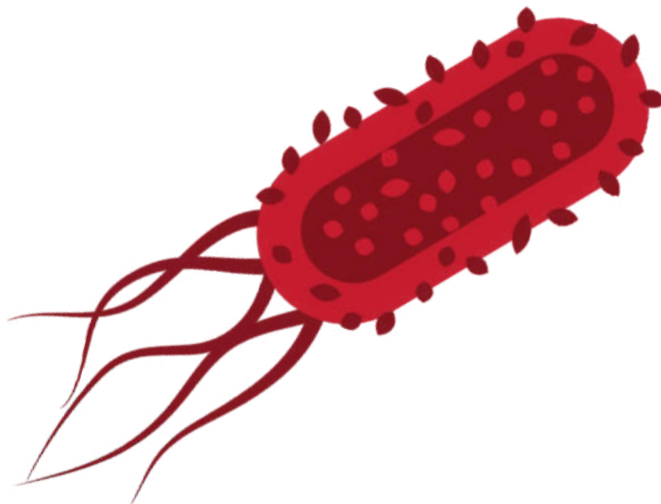
Organisms are classified based on:

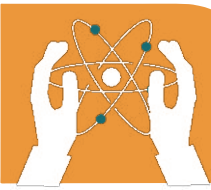
- 1. Cell Structure:** Organisms can be Prokaryotic (without nucleus) or Eukaryotic (with nucleus).
- 2. Mode of Nutrition:** Organisms can be Autotrophic (make their own food) or Heterotrophic (depend on others for food).
- 3. Body Organization:** Organisms can be Unicellular (single-celled) or Multicellular (many-celled).
- 4. Reproduction:** Organisms can perform Asexual (without gametes) or Sexual (with gametes) reproduction.
- 5. Level of Complexity:** Organisms can be simple like single celled bacteria or complex like multicellular plants and animals.



A. Kingdom Monera (Bacteria):

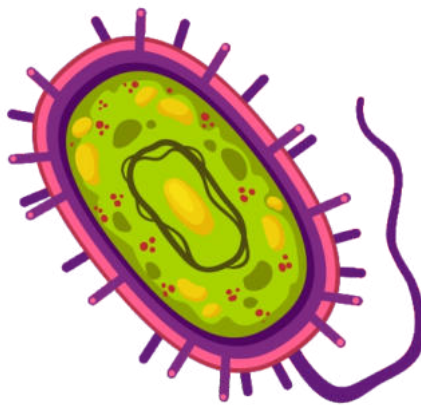
- They are unicellular, microscopic organisms.
- They are Prokaryotic (no nucleus).
- **Examples:** Bacteria, Blue-green algae.





B. Kingdom Protista (Single-celled Eukaryotes):

- They are unicellular, eukaryotic organisms.
- They live in water or moist environments.
- **Examples:** Amoeba, Paramecium, Euglena.



C. Kingdom Fungi (Molds, Mushrooms, Yeasts):

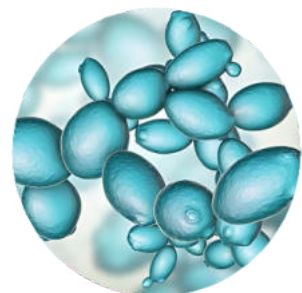
- They are eukaryotic, mostly multicellular (except yeast).
- They usually depend on dead organisms for food.
- **Examples:** Mushrooms, Molds, Yeast.



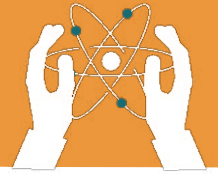
molds



mushrooms



yeasts



D. Kingdom Plantae (Plants):

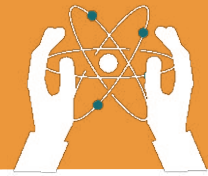
- They are multicellular, eukaryotic.
- They have cell walls made of cellulose.
- **Examples:** Trees, Flowers, Grasses.



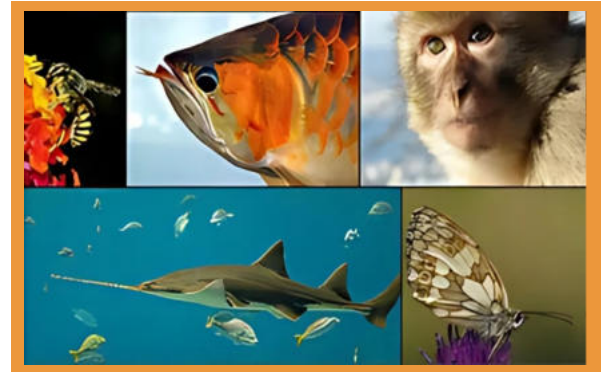
kingdom plantae

E. Kingdom Animalia (Animals):

- They are multicellular, eukaryotic.
- They can move from place to place (motile).
- **Examples:** Humans, Birds, Fish, Insects.



LIVING THINGS



kingdom animalia

1.5 Adaptations in Living Organisms

Living organisms survive in different environments by developing unique characteristics called **adaptations**. Adaptations help them obtain food, reproduce, protect themselves, and adjust to their surroundings. These adaptations can be

A. Structural (physical) adaptations

B. Behavioural adaptations

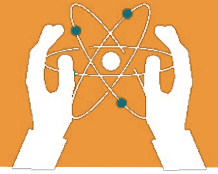
C. Physiological adaptations

A. Structural Adaptations:

These are physical features of an organism that help it survive.

Examples:

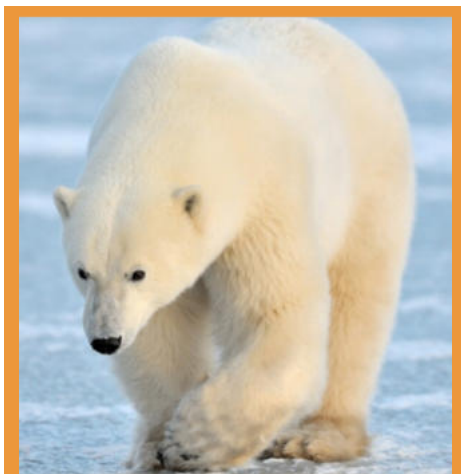
- **Camouflage:** The chameleon changes colour to blend into its surroundings.



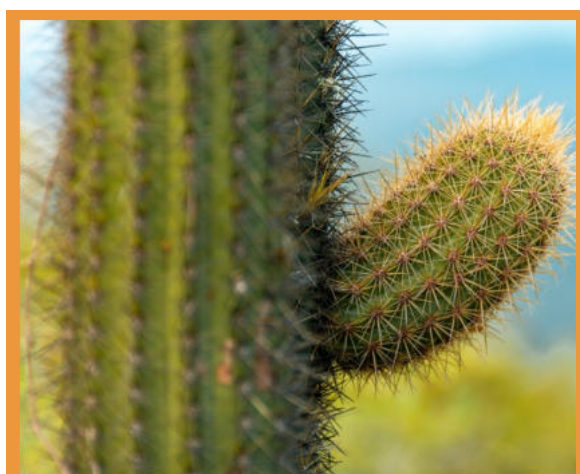
chameleon

• **Body Coverings:**

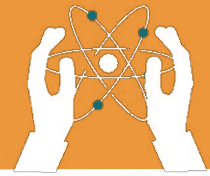
- Polar bears have thick fur to stay warm in the Arctic region.
- Cactus plants have spines instead of leaves to reduce water loss.



polar bear



cactus

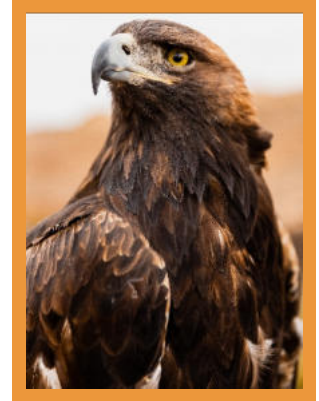


LIVING THINGS

- **Beak Shapes:** Birds have different beak shapes based on their food sources. (e.g., eagles have sharp, curved beaks to tear meat, while hummingbirds have long slender beaks for nectar.)



hummingbird



eagle

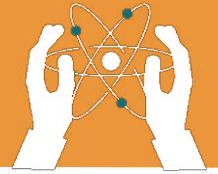
B. Behavioural Adaptations:

These are actions or behaviours that help organisms survive.

Examples:

- **Migration:** Birds like Arctic terns travel long distances to find food and warmer climates.





- **Hibernation:** Bears sleep through the winter to conserve energy.
- **Hunting in Packs:** Wolves hunt in groups to catch large prey.

C. Physiological Adaptations:

These involve internal body processes that help organisms survive.

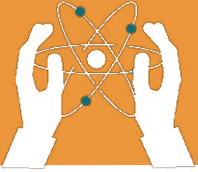
Examples:

- **Venom Production:** Snakes produce venom to catch prey and defend themselves.



venomous snake

- **Food Storage:** Desert animals, like camels, store fat in their humps to survive long periods without food or water.



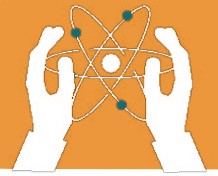
- **Salt Tolerance:** Some plants, like mangroves, filter out excess salt from seawater.



1.6 Adaptations in Different Habitats

A. Desert Adaptations:

- Animals like camels store fat and have long eyelashes to protect against sandstorms.



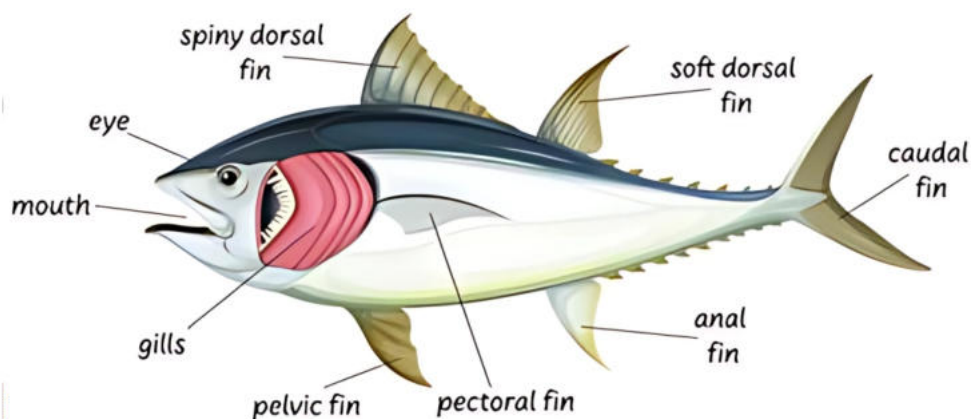
- Cactus store water in thick stems and have deep roots to absorb underground moisture.

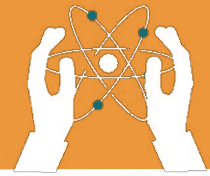


Long eyelashes of camel

B. Aquatic Adaptations:

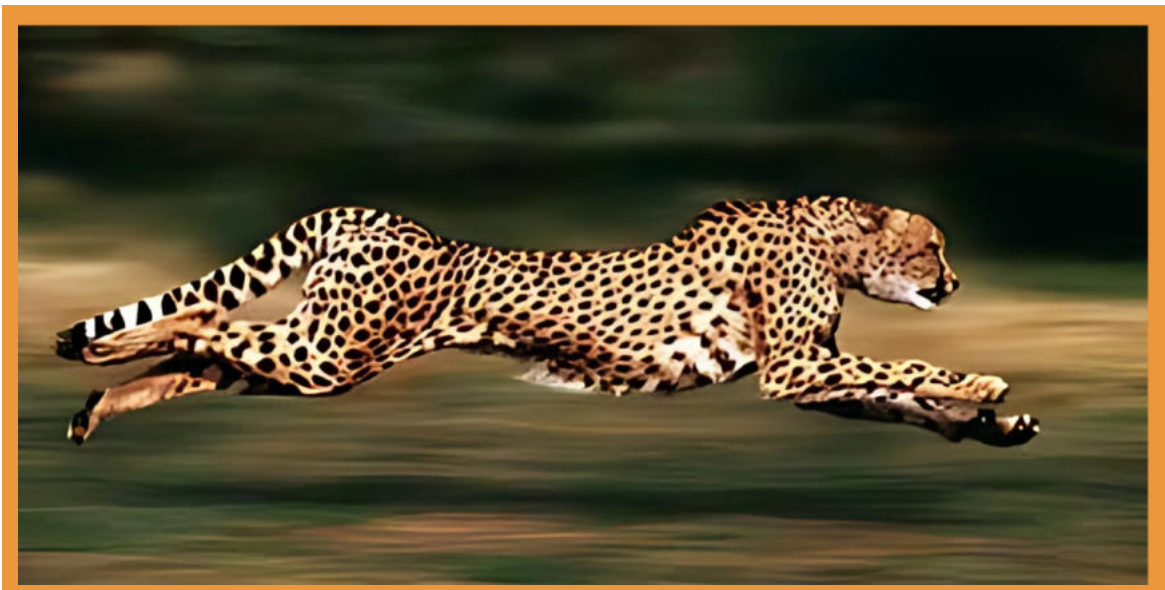
- Fish have gills to extract oxygen from water.
- Whales and dolphins have a layer of fat called blubber to stay warm in cold waters.





C. Grassland Adaptations:

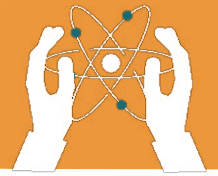
- Cheetahs have a streamlined body for speed to catch prey.
- Grass plants have deep roots to survive fires and drought.



1.7 Human Adaptations

Humans have also adapted to different environments:

- People in cold regions wear thick clothes and build insulated houses.
- People living in high-altitude areas, like Tibet, have larger lungs to take in more oxygen.
- In hot regions, people wear light clothes and drink more water to stay cool.



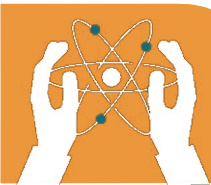
Role of Living Organisms in the Environment

A. Producers (Plants and Algae) – Oxygen and Food Providers:

- Plants and algae perform **photosynthesis**, producing oxygen necessary for all living beings.
- They form the base of the **food chain**, providing energy for herbivores and carnivores.
- Trees absorb carbon dioxide, reducing global warming.

B. Consumers (Animals) – Maintaining Ecological Balance:

- **Herbivores (e.g., deer, cows):** They control plant population and prevent overgrowth.
- **Carnivores (e.g., lions, eagles):** They regulate herbivore numbers, preventing overgrazing.
- **Omnivores (e.g., humans, bears):** They maintain balance by consuming both plants and animals.
- **Pollinators (e.g., bees, butterflies, birds):** They help in plant reproduction by spreading pollen.



C. Decomposers (Fungi, Bacteria) – Recycling Nutrients:

- Decomposers break down dead organisms, returning nutrients to the soil.
- They help in natural waste decomposition, reducing pollution.
- They improve soil fertility for plant growth.

❖ Importance of Adaptations:

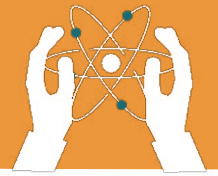
- It helps organisms survive and reproduce.
- It reduces competition for food and space.
- It helps to maintain biodiversity and ecological balance.

DO YOU KNOW?

A **rainforest** is a dense and lush forest that receives a high amount of rainfall throughout the year. These forests are home to more than half of the world's plant and animal species. Found mainly near the equator, such as the **Amazon Rainforest**, **Congo Rainforest**, and **Southeast Asian Rainforests**, they have tall trees, thick vegetation, and multiple layers of plant life.

Rainforests play a crucial role in maintaining Earth's climate by absorbing carbon dioxide and producing oxygen. Despite covering only about **6% of Earth's surface**, they generate **around 20% of the world's oxygen**. These forests are often called the "**lungs of the Earth**" due to their ability to purify the air and regulate global temperatures.





EXERCISES

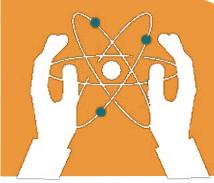
Fill in the blanks.

1. Cells are the _____ units of life.
2. Plants make their food through the process of _____.
3. The _____ system supports the body's structure.
4. All living organisms show _____, which means an increase in size.

Tick(✓) for True and cross (x) for False statements

1. All living organisms grow and develop.
2. Only animals can move from one place to another.
3. Respiration is the process by which living organisms produce energy.
4. Excretion and egestion are the same processes.
5. All living organisms respond to stimuli in their environment.

☐
☐
☐
☐
☐



Choose the correct answer

1. The smallest unit of life is:

- a) Tissue
- b) Cell
- c) Organ
- d) Organ System

2. Which of the following is NOT a characteristic of living organisms?

- a) Growth
- b) Respiration
- c) Decay
- d) Reproduction

3. Living organisms obtain energy through the process of:

- a) Photosynthesis
- b) Excretion
- c) Respiration
- d) Both a and c

4. Which characteristic allows living organisms to respond to changes in their environment?

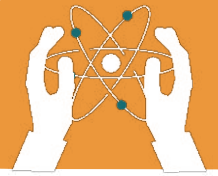
- a) Growth
- b) Nutrition
- c) Adaptations
- d) Movement

5. The process by which living organisms remove waste materials from their body is called:

- a) Excretion
- b) Digestion
- c) Reproduction
- d) Circulation

6. All living things are made up of:

- a) Water
- b) Air
- c) Cells
- d) Rocks



7. Which of the following is an example of a behavioral adaptation?

- a) A chameleon changing its colour to blend into its surroundings.
- b) The development of thick fur in arctic foxes to insulate against cold temperatures.
- c) The migration of monarch butterflies to warmer regions during winter.
- d) The presence of webbed feet in ducks to aid in swimming.

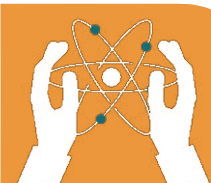
8. What is the primary purpose of camouflage in animals?

- a) To attract mates.
- b) To regulate body temperature.
- c) To avoid predators by blending into the environment.
- d) To assist in hunting by increasing speed.

Answer the Following Questions:

Short-Answer Questions

- 1. What is a cell?**
- 2. Describe three characteristics of living things?**
- 3. Define structural adaptations.**
- 4. How many types of cells are there? Write the difference between unicellular and multicellular organisms.**
- 5. How many kingdoms are there in the classification of living organism?**
- 6. Write the difference between aquatic and desert adaptation?**

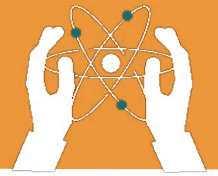


Long-Answer Questions

1. Discuss in detail about classifications of Living Organisms with Examples.
2. What are the main roles of producers, consumers and decomposers in an ecosystem.

Match the Column:

COLUMN A	COLUMN B
Movement	Leaves turning towards light
Respiration	Removal of waste
Excretion	Breaking down food for energy
Growth	Increase in size
Sensitivity	Running of animals



Adaptation Analysis



ACTIVITY

Compare the physical features of animals living in different habitats (e.g., desert vs. rainforest).

Question: How do specific adaptations (like thick fur or webbed feet) help organisms survive in their environments?



Critical Thinking

Cacti are commonly found in deserts, while water lilies thrive in aquatic environments. Compare and contrast the adaptations of cacti and water lilies that enable them to survive in such different habitats. How do these adaptations help them manage water availability?